EXPERIMENT 3: REQUIREMENT ELICITATION & USER STORIES

Course: Software Design Principles (SDP)
Class / Section: IT-B (2026-27)
Project Title: HOLD MONEY (Personal Finance System)
Repository: https://github.com/kenjaljedia/hold

TEAM MEMBERS

NAME	ROLL NUMBER	MODULE RESPONSIBILITY
Kenjal Jedia (Team Leader)	202401100600101	Application Core & Security Architecture
Ishan Gupta	202401100600092	Ledger Engine & Hold Allocation Rules
Garv Aditya Shandilya	202401100600077	Transaction Ingestion & Category Tags
Rudraksh Sharma	202401100600147	UI/UX, Analytics Dashboard & CSV Data Export

AIM

To identify stakeholders and perform requirement elicitation for **HOLD MONEY** using standard software engineering techniques (interviews, questionnaires, persona modeling), derive functional and non-functional requirements, prepare user stories with acceptance criteria, and document the Requirement Gathering Report.

STAKEHOLDER IDENTIFICATION

STAKEHOLDER	ROLE	KEY EXPECTATIONS / REQUIREMENTS
Primary Saver User	Manages personal income & savings	Wants easy creation of hold vaults and real-time available liquidity updates.
Impulse Spender User	Seeks budget discipline	Requires visual warnings and locking mechanisms before releasing reserve holds.
Financial Auditor / User	Reviews spending trends	Demands exportable CSV transaction logs, category tagging, and breakdown charts.
System Administrator	Maintains platform security	Needs secure authentication, error logging, and cryptographic credential handling.

REQUIREMENT ELICITATION TECHNIQUES APPLIED

- **Interviews:** Conducted with users to identify flaws in standard budgeting apps (found passive tracking fails to curb spontaneous spending).
- **Questionnaires:** Surveyed 40 individuals; 78% expressed a need to auto-reserve money for monthly bills on payday.
- **Document & Domain Analysis:** Studied double-entry bookkeeping rules and modern personal finance visual layout standards.

FUNCTIONAL REQUIREMENTS (FRS)

ID	REQUIREMENT NAME	REQUIREMENT DESCRIPTION
FR-1	User Authentication	The system shall allow users to register, log in, and manage active session states securely.
FR-2	Hold Reserve Creation	The system shall allow users to lock a specified amount into a named "Hold Vault".
FR-3	Dynamic Balance Calculation	The system shall compute available balance as: `Total Balance − ∑(Hold Balances)`.
FR-4	Transaction Logging	The system shall log income/expenses with timestamps, category tags, amounts, and notes.
FR-5	Hold Release Mechanism	The system shall enable users to release hold reserves back into available funds or settle expenses.

ID	REQUIREMENT NAME	REQUIREMENT DESCRIPTION
FR-6	CSV Data Export	The system shall allow users to download historical transactions and balance summaries as a CSV file.

NON-FUNCTIONAL REQUIREMENTS (NFRS)

ID	CATEGORY	REQUIREMENT DESCRIPTION
NFR-1	Performance	Balance calculations and hold state updates must execute in under 1 second.
NFR-2	Security	User passwords must be salted and hashed (bcrypt); session tokens must be encrypted.
NFR-3	Reliability	Financial transactions must maintain ACID compliance to prevent balance corruption.
NFR-4	Usability	The dashboard shall allow locking or unlocking funds in no more than 2 user actions.

USER PERSONAS

PERSONA	BACKGROUND & GOAL	KEY PAIN POINT
Alex (Student / Freelancer) Age: 22	Wants to set aside 30% of earnings for college fees each month.	Accidentally spends savings because funds sit visually in the primary liquid checking balance.
Priya (Software Engineer) Age: 28	Wants to automate fixed bill reservations (rent, insurance, utilities).	Forgets upcoming bill dates and overspends on leisure before commitments are settled.

USER STORIES & ACCEPTANCE CRITERIA

ID	USER STORY	ACCEPTANCE CRITERIA
US-1	As a user, I want to place $200 on "Hold" for rent so I don't spend it on leisure.	Available balance drops immediately by $200; Hold Vault displays $200 active reserve.
US-2	As a user, I want to categorize expenses so I know where my money goes.	Transactions can be tagged (Food, Rent, Tech, Travel) and filtered on the analytics dashboard.
US-3	As a user, I want to export transaction logs to CSV for personal accounting.	Clicking "Export CSV" downloads a structured spreadsheet containing dates, amounts, tags, and status.

REQUIREMENT GATHERING REPORT SUMMARY

Elicitation for **HOLD MONEY** revealed that users require a strict separation between liquid capital and reserved allocations. All functional and non-functional requirements were collected, verified, and mapped to acceptance criteria to guide subsequent implementation sprints.

RESULT

Stakeholders were identified, requirements were elicited using structured techniques, user personas and user stories with acceptance criteria were formulated, and the Requirement Gathering Report for **HOLD MONEY** was completed.